

PRAKHAR SHEKHAR PARTHASARTHI

Senior Software Engineer — Java · Distributed Systems · AWS
BEING AN ACCOUNT OF THREE SEASONS AT AMAZON TRAVEL



prakhar.mnnit.2022@gmail.com · github.com/praxstack · linkedin.com/in/prakharshekhar · Shahjahanpur, UP ·
open to relocate · Open to Sr SWE roles

Wherein are recounted the principal achievements of a Java-first backend engineer presently shipping pricing, checkout, and payment infrastructure for *Amazon Travel* — hotels, flights, and insurance. Partial to *CompletableFuture*, idempotent webhooks, small blast radii, and runbooks written before the incident. Three seasons completed; a fourth solicited.

P90 LATENCY

-47%
1.10s → 0.59s on pricing fan-out

CONVERSION

+33%
traveler booking funnel

COST / TXN

-43BPS
idempotent payment pipeline

ON-CALL

20+ → 5
pages / shift · 150+ incidents resolved

Chapter the First

SDE II — AMAZON TRAVEL



AMAZON TRAVEL — Hotels · Flights · Insurance

JUL 2023 — SEP 2025 · BENGALURU

Owner of four Java microservices across the booking funnel — pricing, checkout, payment webhooks, insurance attach.

- I. On the matter of latency. Re-architected pricing fan-out with `CompletableFuture.allOf` and bulkhead `ExecutorService` pools, isolating slow suppliers from the hot path. P90 1.10s → 0.59s (-47%); traveler conversion +33%.
- II. Of payments & idempotence. Idempotent payment webhook on `SQS + Step Functions` with DynamoDB dedupe keys — cost-per-transaction -43 bps and the double-capture incident class extinguished.
- III. A matter of infrastructure. Rewrote six legacy pipelines into `Java CDK` with versioned stacks and blue/green cutovers; rollback rate halved and MTTR reduced by one day.
- IV. Of stewardship. Pruned on-call load from 20+ to ~5 pages per shift; config-hygiene violations 5-6 → 0; resolved 150+ incidents; stopped an estimated ~30 bps funnel-churn on insurance attach.

Chapter the Second

PREVIOUS CHAPTERS



SDE I — AMAZON TRAVEL

JUL 2022 — JUL 2023 · BENGALURU

Built REST endpoints and `@Repository` DynamoDB DAOs wired with **Spring DI**; shipped auto-insurance renewal contributing **+60% renewal completion**. Held the line at **80%+ JUnit coverage** with **JaCoCo** enforced in CI.

SDE INTERN — AMAZON TRAVEL

MAY 2021 — FEB 2022 · BENGALURU

Led the **GraphQL → Java** migration of a checkout surface; delivered the One-Click Renewal flow with **Guice** DI and an end-to-end harness at **90%+ coverage**.



THE APPARATUS



A.	<i>JVM Core</i>	Java 17 · Spring Boot · Spring MVC · Spring DI · Guice · Maven · Gradle	B.	<i>Concurrency</i>	ExecutorService · CompletableFuture · ConcurrentHashMap · ScheduledExecutor · bulkheads
C.	<i>AWS</i>	Lambda · DynamoDB · SQS · Step Functions · API Gateway · AppSync · CloudWatch · ECS · CDK (Java)	D.	<i>Quality</i>	JUnit 5 · Mockito · JaCoCo · integration harnesses · contract tests
E.	<i>Front-End</i>	React (SSR & CSR) · Redux · TypeScript · Vitest	F.	<i>Data Patterns</i>	DynamoDB single-table · idempotency keys · dedupe queues
G.	<i>Ops</i>	runbooks · alarm hygiene · SEV pages · on-call rotations	H.	<i>Method</i>	trunk-based · code-review discipline · postmortems · SLO-first



SUNDRY DEVICES



REDIS SERVER — IN JAVA

github.com/praxstack/redis-server-java

From-scratch **RESP2** protocol with event-loop I/O and a hashmap engine. Written for sport & study; benchmarked against the C implementation to keep one honest about what the JVM can actually do.

SET 154k rps GET 195k rps INCR 128k rps 42 tests
~78% of C

MARKDOWN VIEWER APP

github.com/praxstack/markdown-viewer-app

Zero-config Markdown reader with a plugin architecture and a keyboard-first command palette. Built to fit in a browser tab and stay there — dependency-free at runtime, tested obsessively.

155+ Vitest tests <100KB gzipped 0 runtime deps



LETTERS OF REFERENCE

*of formal instruction*

M.C.A. — Computer Science · MNNIT · 2019–2022 · GPA 9.25/10 Prayagraj

B.SC. (HONS) — Computer Science · Bhaskaracharya College · 2015–2018 · Delhi University

of recognitions received

★ **Amazon Pay Merchants** — Categories Champion, 2023

★ **100+ trust-buster bugs** resolved across travel-funnel quality sprints

